Lab 3: Buffer Overflow and Improper Memory Handling

Name:
Date:
Course:
Team Number:

Section 1 - Lab Overview

1. What is the purpose of this lab?
2. What is a buffer overflow?
3. Why are buffer overflows considered dangerous vulnerabilities?
4. Why was Python used for this lab instead of C or C++?

Section 2 - Program Startup and Initial Observation

5. What message is displayed when the program starts?
6. List the five menu options shown by the program.
    1.
    2.
    3.
    4.
    5.
7. What is the purpose of the simulated stack frame in this lab?

Section 3 - Option 1: Show Fresh Memory Layout

8. After selecting Option 1, what values are shown for the buffer, canary, and return address?

Buffer:
Canary:
Return Address:

9. Why is the buffer initially filled with placeholder values?
10. Why are the canary and return address placed after the buffer?

Section 4 - Option 2: Safe Copy

11. Record the updated memory values after the safe copy.

Buffer:
Canary:

Return Address:
12. Why were the canary and return address not modified?
13. Enter oversized input using Option 2. What happens to the extra characters?

Section 5 - Option 3: Vulnerable Copy

14. Record the updated memory values after the vulnerable copy.
Buffer:
Canary:
Return Address:
15. Which part of the input overwrote the canary?
16. Which part of the input overwrote the return address?
17. Why is this considered insecure behavior?

Section 6 - Option 4: Example Overflow Demo

18. What predefined input string is used in the example overflow demonstration?
19. Why is this example useful for learning about buffer overflows?
20. What memory values changed during the demonstration?

Section 7 - Secure Software Principles

21. Which security principle was violated? (Circle all that apply)

A) Least Privilege　　B) Defense in Depth
C) Secure by Design　　D) Input Validation

22. Explain why that principle applies:

Section 8 - Reflection and Conclusion

23. What is the biggest risk of a real-world buffer overflow attack?
24. What was the most important concept you learned from this lab?
25. Write a short conclusion summarizing what this lab demonstrated and how buffer overflows can be mitigated.

Section 9 - Evidence (Screenshots)

Students must include:

- Screenshot of fresh memory layout
- Screenshot of Safe Copy with normal input
- Screenshot of Vulnerable Copy with overflow input
- Screenshot of Example Overflow Demo
- Screenshot of secure code implementing bounds checking